THE PROGRESS OF CONSOLIDATION IN A CLAY LAYER INCREASING IN THICKNESS WITH TIME

by

R. E. GIBSON

SYNOPSIS

The Paper deals with the one-dimensional consolidation of a clay layer which increases in thickness with time. The pore-water pressures are generated by the increasing weight of superincumbent material and the length of the drainage path varies. Because of these unusual features the standard methods of analysis are inapplicable except in the special instance when the thickness of the layer is proportional to $t^{\frac{1}{2}}$.

It is shown that in general the problem can be reduced to a linear integral equation, a solution of which is obtained for the case of deposition at a constant rate. In conclusion an example is given of a numerical procedure for estimating the progress of consolidation in a rolled fill dam during construction.

Cet article traite de la consolidation en une dimension d'une couche argileuse dont l'épaisseur croit en fonction du temps. L'accroissement de la charge, dû au poids de matériel qui se superpose, engendre les pressions interstitielles, et concurremment la longueur du parcours de drainage varie. A cause de ces caractéristiques spéciales, les méthodes ordinaires ne peuvent être appliquées sauf dans le cas particulier ou l'épaisseur de la couche est proportionelle à $t^{\frac{1}{2}}$.

Il est démontré qu'un général le problème peut être réduit à une équation intégrale linéaire, et une solution est obtenue pour le cas d'une dépôt a taux constant. En conclusion, un exemple expose un procédé numérique pour évaluer la consolidation progressive durant la construction dans un remblai roulé d'un barrage.

INTRODUCTION

It is occasionally required to estimate the progress of consolidation in a clay layer which is increasing in thickness with time. Some typical examples are: the dissipation of pore-water pressure in an earth dam during construction, and the consolidation of a sediment which is accreting at some known rate. When the deposition is fairly uniform over an area whose dimensions are large compared with the thickness of the layer, the consolidation will be approximately one-dimensional and we shall concern ourselves here only with this case.

Some consideration to problems of this nature has already been given by Terzaghi (1925), Ortenblad (1930), Terzaghi and Fröhlich (1936), and R. Gran Olsson (1949, 1953), who developed some ingenious approximate solutions in the case of a constant rate of deposition. In the mathematical treatment it is necessary to satisfy a condition on the pore-water pressure at a *moving* boundary, and for this reason exact* solutions are difficult to obtain. Only one has so far appeared in the literature (Benfield, 1950) and this relates to accretion of clay at a uniform rate on an existing deep layer of similar material which is initially fully consolidated under its own weight.†

While working on problems of this type exact analytical solutions were found for two different rates of deposition. One of these corresponded to the case of sedimentation at a constant rate. The Author believes these solutions to be of interest since it appears unlikely that closed solutions can be obtained for arbitrary rates of deposition.

However, practical problems will frequently involve a time-thickness relation which cannot be represented even approximately by either of those considered in the first part of this Paper. It is then necessary to investigate the progress of consolidation in the layer by some numerical method and, in conclusion, an example of this type of procedure as applied to an actual problem is given.

* Exact within the framework of the usual assumptions made in the theory of one-dimensional consolidation.

† This is the interpretation in terms of the consolidation process of the cosmological problem considered by Benfield.

THE GOVERNING EQUATION

The physical assumptions upon which the following analysis is based are those usually adopted in the theory of one-dimensional consolidation. For ease of presentation it is helpful to consider at this stage a general problem, special cases of which will be studied later.

Sedimentation takes place through still water of depth $H(t)$ to bedrock, the current thickness of the deposit being $h(t)$. (See Fig. 1.) Both $H(t)$ and $h(t)$ are specified functions of the elapsed time t since deposition commenced, and the initial thickness $h(0)$ of the layer is taken as zero.

If one considers the net rate of accumulation of pore-water in a unit volume of the fully saturated sediment at an elevation x above the base of the layer, the continuity equation is readily derived in the form :

$$c_v \frac{\partial^2 p_w}{\partial x^2} = -\frac{\partial \sigma'_{xx}}{\partial t} \qquad \ldots\ldots\ldots \quad (1)$$

where p_w is the water pressure (above atmospheric pressure), and σ'_{xx} is the vertical effective pressure. It has been assumed here that the permeability and compressibility are constant, and that the pore-water is incompressible.

Now the total vertical pressure :

$$\sigma_{xx} = \gamma(h - x) + \gamma_w(H - h) = \sigma'_{xx} + p_w \qquad \ldots\ldots \quad (2)$$

where γ and γ_w are respectively the bulk density of the sediment (assumed constant) and the density of water.* It follows from (1) and (2) that the pore-water pressure p_w in the sediment is governed by the equation :

$$c_v \frac{\partial^2 p_w}{\partial x^2} = \frac{\partial p_w}{\partial t} - \left(\gamma_w \frac{dH}{dt} + \gamma' \frac{dh}{dt}\right) \qquad \ldots\ldots \quad (3)$$

where $\gamma' = \gamma - \gamma_w$. The water pressure at the boundary is :

$$p_w = \gamma_w (H - h)$$

while the condition at $x = 0$ will depend on the nature of the underlying stratum. For example, if this stratum is impermeable $\partial p_w/\partial x = -\gamma_w$ at $x = 0$.

Some simplification results by introducing an *excess* pore-water pressure :

$$u = p_w - \gamma_w H + \gamma_w x.$$

The equation for u is then :

$$c_v \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} - \gamma' \frac{dh}{dt} \qquad \ldots\ldots\ldots \quad (4)$$

and :

$$u = 0, \quad x = h \qquad \ldots\ldots\ldots \quad (5)$$

$$\frac{\partial u}{\partial x} = 0, \quad x = 0 \qquad \ldots\ldots\ldots \quad (6)$$

It will be noted that by working in terms of the excess pore-water pressure the governing equation no longer involves $H(t)$. This is particularly advantageous for, as will be seen later, sedimentation problems (H = constant) can be treated in parallel with bank construction problems where $H(t) = h(t)$.

* The concentration of soil grains in the water is supposedly so small that the density of the water and suspension may be taken as γ_w.

The term $-\gamma' \dfrac{dh}{dt}$ in equation (4) arises because the total vertical pressure σ_{xx} varies with time, this variation being caused by the increasing weight of superincumbent sediment; it would be equal to the rate of increase of the pore pressure if no dissipation took place.

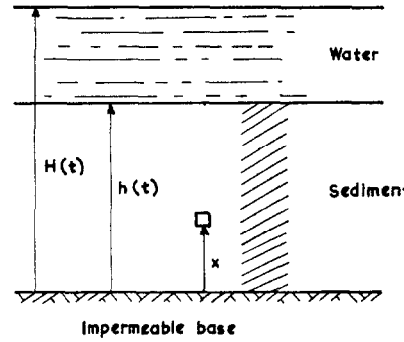


Fig. 1.

SEDIMENTATION ON AN IMPERMEABLE BASE

1. *Thickness of deposit proportional to* $t^{\frac{1}{2}}$

To determine the distribution of pore-water pressure through an accreting deposit entails finding a solution to (4) consistent with the boundary conditions (5) and (6). It is found that the most simple type of solution arises for a rate of sedimentation corresponding to :

$$h = kt^{\frac{1}{2}}$$

and will be examined first.

Replacing x by a new independent variable $\xi = x/h(t)$ and u by a new dependent variable $v = u - \gamma' h(1 - \xi)$ in equation (4) it is found that :

$$c_v \frac{\partial^2 v}{\partial \xi^2} + \xi h \frac{dh}{dt} \frac{\partial v}{\partial \xi} = h^2 \frac{\partial v}{\partial t}.$$

If one seeks a solution to this equation of the form :

$$v = F(\xi)\phi(t)$$

it will be discovered that this is possible only if $h = kt^{\frac{1}{2}}$ and $\phi(t) = t^{\frac{1}{2}}$. The function $F(\xi)$ is determined from the equation :

$$\frac{d^2F}{d\xi^2} + \frac{k^2}{2c_v}\left(\xi \frac{dF}{d\xi} - F\right) = 0$$

of which the complete solution is :

$$F(\xi) = A\left[\exp - \frac{k^2\xi^2}{4c_v} + \frac{k\xi}{2}\left(\frac{\pi}{c_v}\right)^{\frac{1}{2}} \operatorname{erf} \frac{k\xi}{2\sqrt{c_v}}\right] + B\xi$$

where A and B are constants. When these constants have been determined from the boundary conditions (5) and (6) the complete solution is found to be :

$$u = \gamma' k t^{\frac{1}{2}}\left[1 - \left\{\frac{\exp - \dfrac{x^2}{4c_v t} + \dfrac{x}{2}\left(\dfrac{\pi}{c_v t}\right)^{\frac{1}{2}} \operatorname{erf} \dfrac{x}{2\sqrt{c_v t}}}{\exp - \dfrac{k^2}{4c_v} + \dfrac{k}{2}\left(\dfrac{\pi}{c_v}\right)^{\frac{1}{2}} \operatorname{erf} \dfrac{k}{2\sqrt{c_v}}}\right\}\right] \quad . \quad . \quad . \quad (7)$$

Numerical values of excess pore-water pressure have been evaluated from this expression, and curves of $u/\gamma' h$ against x/h are given in Fig. 2 for various values of $k/2\sqrt{c_v}$. It is of interest to note that for this particular rate of deposition the distribution of excess pore pressure through the thickness of the deposit is controlled only by the time independent parameter $k/2\sqrt{c_v}$, and it follows, therefore, that the average degree of consolidation of the layer (see equation 11) is independent of the time. This interesting result appears to be a property of the law $h \propto t^{\frac{1}{2}}$ and to be unaffected by the particular boundary conditions involved in the problem (see equation 15).

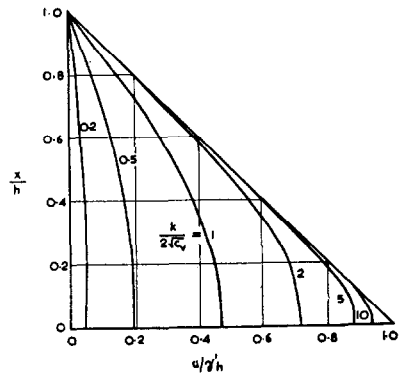


Fig. 2. Relation between $u/\gamma' h$ and x/h for $h = kt^{\frac{1}{2}}$ (impermeable base)

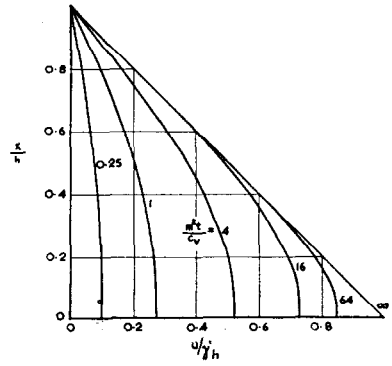


Fig. 3. Relation between $u/\gamma' h$ and x/h for $h = mt$ (impermeable base)

2. *Constant rate of deposition*

We should turn now to the practically important case of constant rate of sedimentation, which is that treated by earlier workers. This simple law might be expected to lead to a simple analytical solution but this unfortunately is not the case.

We choose (Gibson 1958) as the appropriate solution to equation (4) the following expression :

$$u = \gamma' h - t^{-\frac{1}{2}} \int_0^\infty g(\xi)[\exp - (x-\xi)^2/4c_v t + \exp - (x+\xi)^2/4c_v t] d\xi \quad . \quad . \quad (8)$$

which, as may be verified directly, also satisfies the boundary condition (6). The function $g(\xi)$ is arbitrary at this stage, and must now be chosen so that the remaining condition (5) is also satisfied. This requirement is met with if g is determined from the integral equation :

$$\tfrac{1}{2}\gamma' h t^{\frac{1}{2}} \exp \frac{h^2}{4c_v t} = \int_0^\infty g(\xi) \cosh \frac{h\xi}{2c_v t} \exp - \frac{\xi^2}{4c_v t} d\xi.$$

It does not appear that this equation can be solved for an arbitrary function $h(t)$, but we notice that in the special case when :

$$h = mt$$

that part of the integrand involving the hyperbolic function may be absorbed in the unknown function. With changes of variable :

$$\xi^2 = \tau, \quad p = 1/4c_v t \quad \text{and} \quad F(\tau) = g(\tau^{\frac{1}{2}})\tau^{-\frac{1}{2}} \cosh \frac{k\tau^{\frac{1}{2}}}{2c_v}$$

the equation becomes :

$$\tfrac{1}{8} m\gamma' c_v^{-\frac{3}{2}} p^{-\frac{3}{2}} \exp \frac{m^2}{16 c_v^2 p} = \int_0^\infty F(\tau) e^{-p\tau} d\tau.$$

This type of integral equation has been studied extensively, and using known methods the following solution is derived :

$$F(\tau) = \frac{\gamma'}{2\pi^{\frac{1}{2}} c_v^{\frac{1}{2}}} \sinh \frac{m\tau^{\frac{1}{2}}}{2c_v}$$

whence :

$$g(\xi) = \frac{\gamma' \xi}{2\pi^{\frac{1}{2}} c_v^{\frac{1}{2}}} \tanh \frac{m\xi}{2c_v}. \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (9)$$

The excess pore-water pressure is then, from (8) and (9), given by :

$$u = \gamma' mt - \gamma'(\pi c_v t)^{-\frac{1}{2}} \exp \frac{-x^2}{4c_v t} \int_0^\infty \xi \tanh \frac{m\xi}{2c_v} \cosh \frac{x\xi}{2c_v t} \exp - \frac{\xi^2}{4c_v t} d\xi \quad . \quad . \quad (10)$$

This integral cannot be expressed in terms of tabulated functions, but it is of a type which can be readily evaluated numerically (see Goodwin, 1949).

Curves of $u/\gamma' h$ against x/h for various values of the time factor $T = m^2 t/c_v$ are shown in Fig. 3. The variation in the gradient of excess pore-water pressure at the surface of the sediment with time is given in the form of a curve of $\frac{1}{\gamma'}\left(\frac{\partial u}{\partial x}\right)_{x=h}$ against the time factor in Fig. 4.

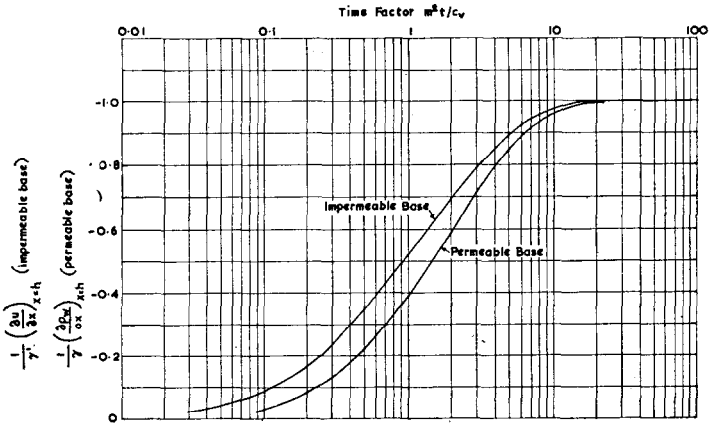


Fig. 4. Variation of undissipated excess pore-water pressure gradient at the surface of layer with time when $h = mt$

Finally, the average degree of consolidation $\overline{U}$ of the deposit, defined by :

$$\overline{U} = 1 - \frac{\int_0^h u \, dx}{\gamma' \int_0^h (h - x) dx} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (11)$$

is given as a function of T in Fig. 5.

It is pertinent to note that the surface gradient and the average degree of consolidation are simply related. By integrating equation (4) with respect to x over the range 0 to h and making use of both boundary conditions the following is obtained :

$$c_v \left(\frac{\partial u}{\partial x}\right)_{x=h} = \frac{d}{dt} \int_0^h u \, dx - \gamma' h \frac{dh}{dt}$$

and by using (11) one finds that :

$$\frac{1}{\gamma'} \left(\frac{\partial u}{\partial x}\right)_{x=h} = - \frac{1}{2c_v} \frac{d}{dt} (h^2 \overline{U})$$

which relation holds for any rate of sedimentation. In the present case with $h = mt$ it reduces to :

$$\frac{1}{\gamma'} \left(\frac{\partial u}{\partial x}\right)_{x=h} = - T\left(\tfrac{1}{2} T \frac{d\overline{U}}{dT} + \overline{U}\right).$$

CONSOLIDATION OF A SEDIMENT SUBSEQUENT TO DEPOSITION

If, in the example treated above, the layer increases in thickness at a uniform rate up to a time t_1 and then deposition ceases, the subsequent progress of consolidation may be predicted using methods well known in the mathematical theory of heat conduction.

At the moment when deposition ceases, the distribution of excess pore-water pressure will be given by the expression (10) with $t = t_1$. This distribution shall be denoted by $u_0(x, t_1)$ and it will form the " initial " condition for the subsequent dissipation which will be controlled by the familiar equation :

$$c_v \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (12)$$

(since $dh/dt = 0$ for $t > t_1$), with boundary conditions :

$$u = 0, \quad x = mt_1$$
$$\frac{\partial u}{\partial x} = 0, \quad x = 0$$

It can then be shown (Carslaw and Jaeger, 1947) that for $t \geqslant t_1$ the pore pressure is given by

$$u = \frac{2}{mt_1} \sum_{n=0}^{\infty} \exp - \frac{\pi^2 c_v (t - t_1)}{4m^2 t_1^2} \cos \frac{(2n+1)\pi x}{2mt_1} \int_0^{mt_1} u_0(\lambda, t_1) \cos \frac{(2n+1)\pi\lambda}{2mt_1} d\lambda \quad . \quad (13)$$

The corresponding expression for the average degree of consolidation can be obtained using (11) and (13), and some typical curves of $\bar{U}$ against T following the end of deposition are shown in Fig. 5. These curves are shown breaking away from the deposition curve at values of $m^2 t_1/c_v = 0{\cdot}4$, 1, 3, and 10. When dissipation ceases at other values of t_1 the corresponding dissipation curves can be sketched in with sufficient accuracy.

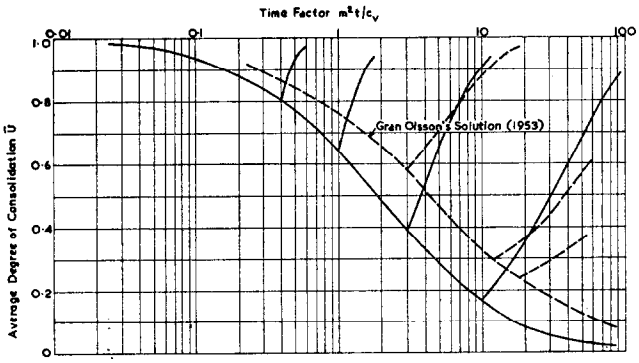


Fig. 5. Relationship between time factor m^2t/c_v and average degree of consolidation $\bar{U}$ during and after formation of clay deposit with impermeable base

Before leaving the sedimentation problem it is instructive to make some comparison with the numerical values obtained by earlier workers. The approximate solution given by Gran Olsson (1953) has been plotted in Fig. 5, and, remembering the severe idealizations implicit in this mathematical treatment, the numerical difference between the two solutions is scarcely of any practical significance.

CONSOLIDATION OF A SATURATED FILL PLACED ON AN IMPERMEABLE BASE

The prediction of the progress of consolidation in a fill of soft saturated clay is closely related to the above problem. Suppose the fill of thickness $h(t)$ to be placed upon an im-

permeable layer. The water level will be at the top of the layer, and this is expressed by the condition $H = h$.

Therefore the equation governing the water pressure will in this case be :

$$c_v \frac{\partial^2 p_w}{\partial x^2} = \frac{\partial p_w}{\partial t} - \gamma \frac{dh}{dt} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (14)$$

Introducing an excess pore-water pressure :

$$u = p_w - (h - x)\gamma_w$$

it is found that :

$$c_v \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} - \gamma' \frac{dh}{dt}$$

as in the sedimentation problem (equation 4), with the same boundary conditions (5) and (6), and the solutions (7) and (10) apply therefore without change to this new situation.

CONSOLIDATION OF A SATURATED FILL ON A PERMEABLE BASE

If the fill is underlain by a permeable stratum which is either dry or in which the water pressure is maintained at atmospheric pressure, then the boundary conditions will be $p_w = 0$ at $x = 0$ and h. It will prove most convenient to work now in terms of the water pressure p_w (equation 14).

The change in the choice of variable, and in the boundary condition on the base of the fill, introduces no important modifications in method of solution for the case when $h = kt^{\frac{1}{2}}$. Below is quoted, for completeness, the solution to this case :

$$\frac{p_w}{\gamma h} = \left(1 - \exp - \frac{x^2}{4c_v t}\right) - \frac{x}{h}\left(1 - \exp - \frac{k^2}{4c_v}\right) + \frac{x}{2}\left(\frac{\pi}{c_v t}\right)^{\frac{1}{2}}\left[\operatorname{erf}\frac{k}{2c_v^{\frac{1}{2}}} - \operatorname{erf}\frac{x}{2c_v^{\frac{1}{2}}t^{\frac{1}{2}}}\right] \quad (15)$$

curves of which are shown in Fig. 6. It is tacitly assumed here that the boundary condition $p_w = 0$ on $x = h$ holds at all times, and that if there is a reversal of flow there is always sufficient rainfall available to prevent the formation of capillary tensions (see, for instance, Terzaghi, 1943, pp. 320–323).

Consider now this same problem for a layer increasing in thickness at a constant rate, and choose a solution to (14) of the form :

$$p_w = \gamma h - \frac{\gamma x}{2\pi^{\frac{1}{2}}c_v^{\frac{1}{2}}}\int_0^t \frac{h(\tau)}{(t-\tau)^{\frac{3}{2}}} \exp - \frac{x^2}{4c_v(t-\tau)}\, d\tau + t^{-\frac{1}{2}}\int_0^\infty f(\xi)\left[\exp - \frac{(x-\xi)^2}{4c_v t} - \exp - \frac{(x+\xi)^2}{4c_v t}\right] d\xi.$$

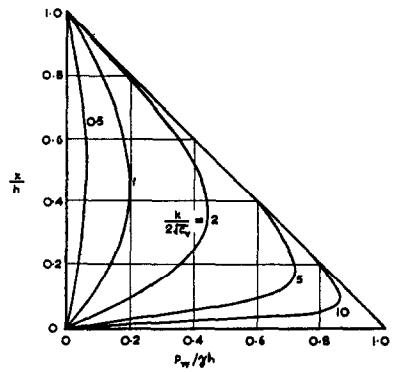


Fig. 6. Relation between $p_w/\gamma h$ and x/h for $h = kt^{\frac{1}{2}}$ (permeable base)

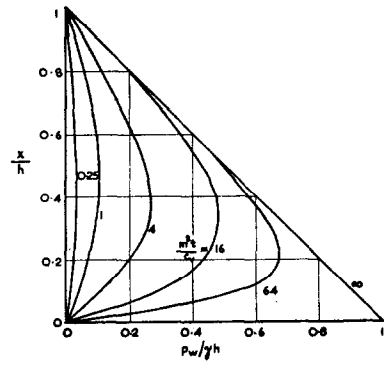


Fig. 7. Relation between $p_w/\gamma h$ and x/h for $h = mt$ (permeable base)

That this solution also satisfies the boundary condition $p_w = 0$ on $x = 0$ may be shown by invoking a lemma credited to Holmgren (Holmgren, 1908) which shows that the second term on the right-hand side of this equation approaches $\gamma h(t)$ as $x \to 0$. The unknown function $f(\xi)$ is now found as the solution to an integral equation as before to be :

$$f(\xi) = \frac{\gamma\xi}{2\pi^{\frac{1}{2}}c_v^{\frac{1}{2}}}\left[1 + \frac{m\xi}{2c_v}\left(1 - \coth\frac{m\xi}{2c_v}\right)\right]$$

and this ensures that $p_w = 0$ at $x = h$. The water pressure distribution is then :

$$p_w = -\gamma x\left(1 + \frac{mx}{2c_v}\right) + \frac{\gamma m}{2c_v}(\pi c_v t)^{-\frac{1}{2}} \exp -\frac{x^2}{4c_v t}\int_0^\infty \xi^2 \coth\frac{m\xi}{2c_v}\sinh\frac{x\xi}{2c_v t}\exp -\frac{\xi^2}{4c_v t}\cdot d\xi.$$

Again, this integral must be evaluated numerically, and the relevant isochrones have been plotted in Fig. 7. The variation of the surface gradient of water pressure with time is also given in Fig. 4.

ESTIMATION OF PORE-WATER PRESSURE SET UP IN AN EARTH DAM DURING CONSTRUCTION

Up to the present the discussion has been restricted to fully saturated clays, but the analysis may be extended to partly saturated soils by introducing the pore-water pressure parameter $\bar{B}$ (Skempton, 1954). This is necessary if it is desired to estimate the pore-water pressures likely to be set up in an earth dam during and after construction. During the early stages of building the pore-water flow is sensibly vertical, and the equation controlling the water pressure in the embankment can be expressed in the form :

$$c_v\frac{\partial^2 p_w}{\partial x^2} = \frac{\partial p_w}{\partial t} - \gamma\bar{B}\frac{dh}{dt} \qquad \ldots\ldots (16)$$

where the parameter $\bar{B}$ is the fraction of the water pressure set up at any point (under conditions of no drainage) in the partly saturated soil to the water pressure that would be set up under the same conditions in a fully saturated soil. It should be noted that the coefficient of consolidation c_v in this equation is no longer a simple parameter depending only upon the coefficient of permeability and the soil compressibility.*

The significance of $\bar{B}$ in the above equation can be readily appreciated if one considers a bank built so rapidly that negligible consolidation has taken place by the end of construction. From equation (16) the following is obtained :

$$\frac{\partial p_w}{\partial t} = \gamma\bar{B}\frac{dh}{dt}$$

and hence :

$$p_w = \gamma\bar{B}h + \phi(x)$$

where ϕ is an arbitrary function.

Since $p_w = 0$ when $x = h$, it follows that

$$\phi(h) = -\gamma\bar{B}h$$

whence

$$\phi(x) = -\gamma\bar{B}x.$$

Therefore

$$\bar{B} = p_w/\gamma(h - x)$$

and this is in accord with the usual definition of $\bar{B}$.

* The derivation and discussion of the basis of this equation is outside the scope of this Paper ; reference may be made to Florin (1948), Biot (1941, 1956) for the derivation of the equations governing consolidation of partly saturated soils.

It is of interest to note that the pore-water will subsequently leave the bank from the top surface only if :

$$\left(\frac{\partial p_w}{\partial x}\right)_{x=h} < -\gamma_w$$

that is if

$$\bar{B} > \gamma_w/\gamma.$$

If $\bar{B} < \gamma_w/\gamma$ then water must be supplied to the top of the bank if the condition $p_w = 0$ there is to remain true.*

It will be seen on comparing equations (14) and (16) that all the solutions previously obtained for a fully saturated soil are validated for a partly saturated soil by simply replacing γ by $\bar{B}\gamma$. Nevertheless, it will only be in special circumstances that these solutions will prove of practical utility. In the following section, therefore, the use of a numerical technique is considered which can be applied directly to determine the pore-water pressure distribution corresponding to any specified rate of thickening.

NUMERICAL METHOD

In the (x, t) plane (Fig. 8) the domain under consideration will be bounded by the line

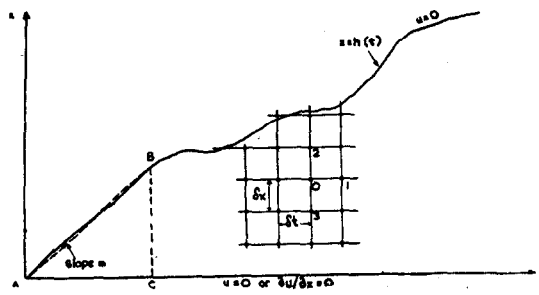


Fig. 8.

$x = 0$ and the curve $x = h(t)$. If this region is divided into equal rectangles of sides δx and δt then at a typical point 0 of the network one may replace the differential equation (14) approximately by the difference equation :†

$$c_v \frac{(p_2 + p_3 - 2p_0)}{(\delta x)^2} \doteq \frac{(p_1 - p_0)}{\delta t} - \gamma \bar{B} \left(\frac{\delta h}{\delta t}\right)_0$$

or :

$$p_1 \doteq p_0 + \beta(p_2 + p_3 - 2p_0) + \gamma \bar{B} \delta h \quad . \quad . \quad . \quad . \quad . \quad (17)$$

where the net parameter $\beta = c_v \delta t/(\delta x)^2$. This equation allows the water pressure distribution at a time $t + \delta t$ to be calculated directly from the distribution at time t. In the more usual type of consolidation problem the solution is carried forward by a step-by-step procedure from a given initial pressure distribution (see, for example, Gibson and Lumb, 1953). But in the problems the reader is concerned with here the layer does not exist initially, and no initial pore-pressure distribution is therefore given. For this reason it would appear that the simple numerical treatment cannot be applied directly.‡ If, however, the solution

* If no free water is available capillary menisci will be formed at the top of the bank and a new boundary condition of zero flow will obtain, namely $\partial p_w/\partial x = -\gamma_w$.

† Dropping, for convenience, the suffix w.

‡ More sophisticated numerical techniques are discussed by Crank, 1956, and some of these can be adapted to overcome this difficulty.

could be carried forward some way from the origin A (Fig. 8) by alternative means then a starting distribution would be found for the numerical step-by-step process. Fortunately, in many practical problems, it is a reasonable approximation to take the initial rate of thickening as constant (line A–B, Fig. 8) and the analytical solutions previously obtained for the case $h = mt$ (Figs 3 and 7) can be used directly to carry the solution forward from A to the line B–C.

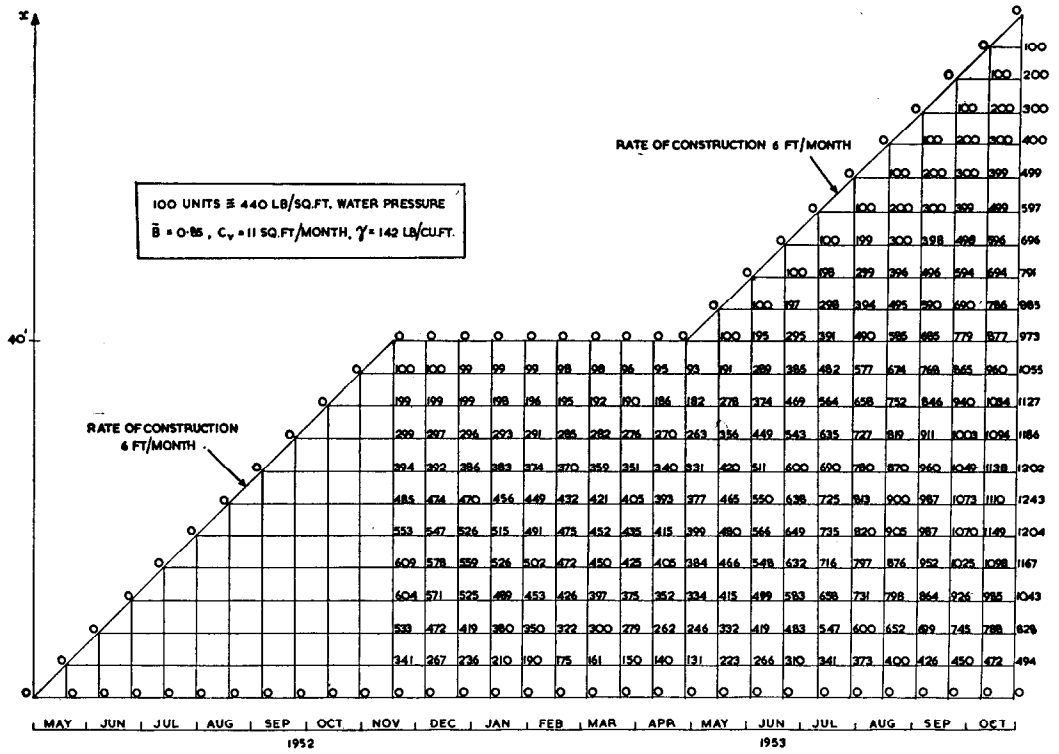


Fig. 9. Predicted pore-water pressure distribution with no drainage blanket

To demonstrate how this technique is applied, in the following section an actual example which embodies the main features of the method is quoted for consideration.

NUMERICAL EXAMPLE

During the initial stages in the construction of a 109-ft-high earth dam in Wales, pore-pressure cells were installed in the bank material (Penman, 1956; and Sheppard and Aylen, 1957). At the end of the first construction season, when nearly 40 ft of fill had been placed, the pore pressures recorded were about 75% of the overburden pressure. After a further 5 months, during which no further construction occurred, these pressures fell only to 60% of the overburden pressure. An analysis was carried out at this stage in an attempt to forecast the pattern of pore-water pressures during the next construction season when it was proposed to place a further 36 ft of fill.

As the dam was underlain by gravel in the valley bottom it was assumed that the water pressure was zero (above atmospheric pressure) at the top and base of the layer. The first layer, 40 ft thick, had been placed in 6·6 months at a fairly uniform rate, and the pore-pressure distribution at the end of this period was therefore determined on the basis of Fig. 7

with $m = \dfrac{40}{6{\cdot}6} = 6$ ft/month, $c_v = 11$ sq. ft/month,* $\bar{B} = 0{\cdot}85$ and $\gamma = 142$ lb/cu. ft.† This estimated distribution of p_w at $t = 6{\cdot}6$ months is shown in Fig. 9 where $\gamma\bar{B}\delta h = 100$ units $= \dfrac{40}{11} \times 0{\cdot}85 \times 142 = 440$ lb/sq. ft water pressure. Using the same values of the parameters the solution was carried forward numerically using equation (17) with $\gamma\bar{B}\delta h = 0$ and $\beta = 0{\cdot}5$, the calculated values being shown in Fig. 9. When construction recommenced at the rate of 5 ft/month the value of $\gamma\bar{B}\delta h$ changed from 0 to 100 units, and the numerical solution was extended to the end of the second lift (see Fig. 9) again using the same values of $\bar{B}$,‡ γ and c_v.

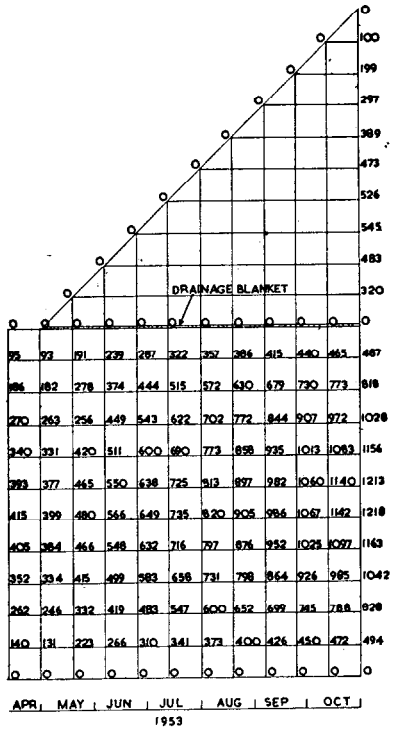


Fig. 10. Predicted pore-pressure distribution with drainage blanket

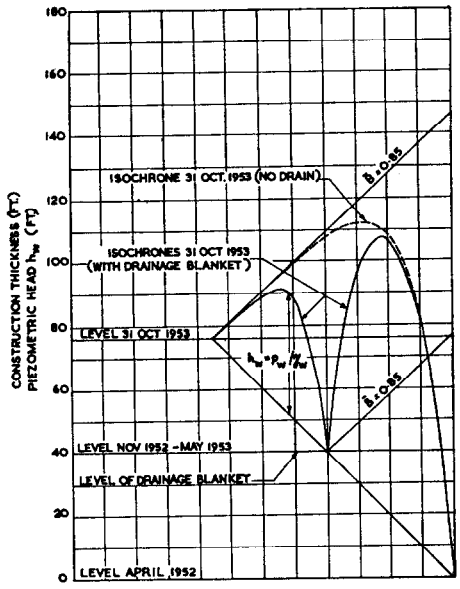


Fig. 11. Forecast of water-pressure distribution October 1953: (a) with no drains; (b) with drainage blanket

These final values of water pressure (Fig. 11) were used in an effective stress stability analysis and the factor of safety was found to be unacceptable (Sheppard and Aylen, 1957). The incorporation of a horizontal drainage blanket between the two layers was therefore considered. The necessary modification was readily allowed for in the numerical work (Fig. 10) and the isochrones at the end of the second lift are shown in Fig. 11. With this distribution of water pressure the factor of safety against sliding was recalculated and found to be adequate.

* This value gave the best fit with the observations up to the beginning of the second construction season (Skempton, 1957).

† Average values of $\bar{B}$ and γ obtained from laboratory and field tests.

‡ Later field evidence showed that $\bar{B}$ associated with further load increments *decreased* with increasing consolidation (Skempton, 1957). This phenomenon has been discussed in detail by Bishop, 1957.

A more detailed comparison between calculated and observed pore pressures than is appropriate here has been given by Sheppard and Aylen (1957), and good agreement was found except towards the end of construction when the lateral flow of pore-water became important.

It must be emphasized, in conclusion, that generally the rôle of theory in a problem of this type is to provide a rough prediction of the water pressures to guide preliminary design and choice of construction rate. However, the parameters $\bar{B}$ and c_v are sensitive to the placement condition and in particular to the water content. With close control therefore it should be possible to achieve a fairly reliable forecast based on the results of laboratory tests. But with unfavourable borrow pit or climatic conditions a close control of the placement water content may not be practicable and in these circumstances the observation of water pressures during construction cannot be dispensed with.

REFERENCES

BENFIELD, A. E., 1950. " The temperature in an accreting medium with heat generation." *Quart. J. appl. Math.*, 7 : 436–439.

BIOT, M. A., 1941. " General theory of three-dimensional consolidation." *J. appl. Phys.*, 12 : 2 : 155.

BIOT, M. A., 1956. " General solutions of the equations of elasticity and consolidation for a porous material." *J. appl. Mech.*, 23 : 1 : 91–96.

BISHOP, A. W., 1957. " Some factors controlling the pore pressures set up during the construction of earth dams." *4th Int. Conf. Soil Mech., London*, 2 : 294–300.

CARSLAW, H. S., *and* J. C. JAEGER, 1947. " Conduction of heat in solids." Clarendon Press, Oxford.

CRANK, J., 1956. " The mathematics of diffusion." Clarendon Press, Oxford.

FLORIN, V. A., 1948. " Teoriya uplotneniya zeml'anih mass." *Moscow.*

GIBSON, R. E., 1958. " A heat conduction problem involving a specified moving boundary." Forthcoming in *Quart. J. appl. Math.*

GIBSON, R. E., *and* P. LUMB., 1953. " Numerical solution of some problems in the consolidation of clay." *Proc. Instn civ. Engrs*, 1 : 2 : 182.

GOODWIN, E. T., 1949. " The evaluation of integrals of the form $\int_{-\infty}^{\infty} f(x)e^{-x^2}dx$." *Proc. Camb. Phil. Soc.*, 45 : 241.

HOLMGREN, E., 1908. *Arkiv für Matematik*, 4 : 18 : 8–12.

OLSSON, R. G., 1949. " Rigorous solution of a differential equation in soil mechanics." *Quart. J. appl. Math.*, 7 : 338–342.

OLSSON, R. G., 1953. " Approximate solution of the progress of consolidation in a sediment." *3rd Int. Conf. Soil Mech., Zurich*, 1 : 38–42.

ORTENBLAD, A., 1930. " Mathematical theory of the process of consolidation of mud deposits." *J. Math. and Phys.*, 9 : 2 : 73–149.

PENMAN, A. D. M., 1956. " A field piezometer apparatus." *Géotechnique*, 6 : 57–65.

SHEPPARD, G. A. R., *and* L. B. AYLEN, 1957. " The Usk scheme for the water supply of Swansea." *Proc. Instn civ. Engrs*, 7 : 246–265.

SKEMPTON, A. W., 1954. " The pore-pressure coefficients *A* and *B*." *Géotechnique*, 4 : 143–147.

SKEMPTON, A. W., 1957. *Discussion* on Paper by G. A. R. Sheppard and L. B. Aylen, " The Usk scheme for water supply of Swansea." *Proc. Instn civ. Engrs*, 7 : 267–269.

TERZAGHI, K., 1925. " Erdbaumechanik auf bodenphysikalischer Grundlage." *Deuticke, Vienna*, p. 175.

TERZAGHI, K., 1943. " Theoretical Soil Mechanics." *Wiley, New York*, pp. 320–323.

TERZAGHI, K., *and* O. K. FRÖHLICH, 1936. " Theorie der Setzung von Tonschichten." *Deuticke, Vienna*, pp. 91–96.